

Views of centres

Ross Street¹

Math-Stat Dept/CoACT, Macquarie University

Category Seminar, National Autonomous University of Mexico

¹New material is joint with Branko Nikolić

Outline

1. An eccentric look at the structures of interest through morphisms.

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4. Davydov's full centre as a centre.

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5. Categories graded by a group G .

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1. An eccentric look at the structures of interest through morphisms.
2. Centres of a category, bicategory, monoidal category, and monoidale (= pseudomonoid) in a braided closed monoidal category.
3. Our application of centre to tangles.
4. Davydov's full centre as a centre.
5. Categories graded by a group G .
6. An example of a higher full centre inspired by the Turaev-Virelizier [JPAA 2013] G -graded version of Müger's Theorem [JPAA 2003] about the centre of a semisimple spherical monoidal $\mathbb{C}$ -linear category being modular.

One-dimensional structures

1. A *right-unital magma* (*ruma*) is a set A equipped with a binary operation $A \times A \rightarrow A$ and an element $i \in A$ satisfying $a \cdot i = a$. If also $i \cdot a = a$, we call it a *unital magma* (*uma*)
2. Any monoid gives an example of a *uma*. In particular, the set $[X, X]$ of endofunctions on a set X is a *ruma* via composition $\circ$ and $i = 1_X$.

3. As expected, a *ruma morphism* $f : A \rightarrow B$ is a function satisfying $f(a) \cdot f(b) = f(a \cdot b)$ and $i = f(i)$.
4. Maybe less expected, a *commorphism* $t : X \rightarrow A$ from a set X to a ruma A is a function satisfying $f(a) \cdot f(b) = f(b) \cdot f(a)$.
5. (Eckmann-Hilton) Suppose A has two magma structures $A \times A \xrightarrow{\cdot} A$ and $A \times A \xrightarrow{\star} A$ with the same two-sided unit i . If $\star$ is a magma morphism for the $\cdot$ structure then the two binary structures agree and the magma is a commutative monoid.

Let $\ell : A \rightarrow [A, A]$ be the function defined by $\ell(a)(b) = a \cdot b$.

Proposition (Exercise!)

Let A be a ruma.

- a. $\ell : A \rightarrow [A, A]$ is a ruma morphism if and only if A is a monoid.*
- ca. $\ell : A \rightarrow [A, A]$ is a commorphism if and only if A is a commutative monoid.*

Note that, in (ca), we are still only assuming A is a ruma, not a monoid. Commutativity and associativity both come out of the commorphism condition.

Two-dimensional structures

1. A *right-unital magmale (rumale)* is a category $\mathcal{A}$ equipped with a functor $\otimes : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$, an object $I \in \mathcal{A}$, and a natural family of isomorphisms $\rho_A : A \otimes I \xrightarrow{\cong} A$.
2. Any monoidal category gives an example. In particular, the category $[\mathcal{X}, \mathcal{X}]$ of endofunctors on a category $\mathcal{X}$ is a rumale via composition $\circ$ and $I = 1_{\mathcal{X}}$.
For any rumale $\mathcal{A}$, we write $L : \mathcal{A} \rightarrow [\mathcal{A}, \mathcal{A}]$ for the functor $LA = A \otimes -$ corresponding to $\mathcal{A} \times \mathcal{A} \xrightarrow{\otimes} \mathcal{A}$.

Rumale morphisms

A *rumale morphism* $F : \mathcal{A} \rightarrow \mathcal{B}$ is a functor equipped with a natural $F_2 : FA \otimes FB \xrightarrow{\cong} F(A \otimes B)$ and an $F_0 : I \xrightarrow{\cong} FI$ such that

$$\begin{array}{ccc} FA \otimes FI & \xrightarrow{F_2} & F(A \otimes I) \\ \uparrow 1 \otimes F_0 & & \downarrow F \rho_A \\ FA \otimes I & \xrightarrow{\rho_{FA}} & FA \end{array}$$

A rumale morphism structure on the functor $L : \mathcal{A} \rightarrow [\mathcal{A}, \mathcal{A}]$ amounts to natural families of isomorphisms

$$\alpha_{A,B,C} : A \otimes (B \otimes C) \xrightarrow{\cong} (A \otimes B) \otimes C \quad \text{and} \quad \lambda_A : A \xrightarrow{\cong} I \otimes A$$

such that the composite

$$A \otimes B \xrightarrow{1 \otimes \lambda_A} A \otimes (I \otimes B) \xrightarrow{\alpha_{A,I,B}} (A \otimes I) \otimes B \xrightarrow{\rho_A \otimes 1} A \otimes B$$

is the identity. Call this a *premonoidal category* when $\rho_I \circ \lambda_I = 1_I$ also.

Strong monoidal functors

A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ between premonoidal categories is *strong monoidal* when it is a rumale morphism such that

$$\begin{array}{ccccc}
 FA \otimes (FB \otimes FC) & \xrightarrow{1 \otimes F_2} & FA \otimes F(B \otimes C) & \xrightarrow{F_2} & F(A \otimes (B \otimes C)) \\
 \alpha \downarrow & & & & \downarrow F\alpha \\
 (FA \otimes FB) \otimes FC & \xrightarrow{F_2 \otimes 1} & F(A \otimes B) \otimes FC & \xrightarrow{F_2} & F((A \otimes B) \otimes C)
 \end{array}$$

$$\begin{array}{ccc}
 FI \otimes FA & \xrightarrow{F_2} & F(I \otimes A) \\
 F_0 \otimes 1 \uparrow & & \uparrow F\lambda_A \\
 I \otimes FA & \xleftarrow{\lambda_{FA}} & FA
 \end{array}$$

- ▶ For a premonoidal category $\mathcal{A}$, a strong monoidal structure on the functor $L : \mathcal{A} \rightarrow [\mathcal{A}, \mathcal{A}]$ amounts to a monoidal structure on the category $\mathcal{A}$.

The Mac Lane pentagon for α comes from the hexagon for L_2 since $[\mathcal{A}, \mathcal{A}]$ is strict monoidal (i.e. α, ρ, λ are identities).

- ▶ For a monoidal category $\mathcal{A}$, a strong monoidal structure on the functor $\otimes : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ amounts to a braiding on $\mathcal{A}$.

The strong monoidal structure gives a middle-of-four isomorphism

$$(A \otimes B) \otimes (C \otimes D) \cong (A \otimes C) \otimes (B \otimes D)$$

which yields the braiding

$$\gamma_{B,C} : B \otimes C \cong C \otimes B$$

on setting $A = D = I$ and using ρ, λ .

Yang-Baxter operators

A *Yang-Baxter operator* on a functor $T : \mathcal{X} \rightarrow \mathcal{A}$ from a category $\mathcal{X}$ to a strict monoidal category $\mathcal{A}$ is a natural family of isomorphisms

$$y_{A,B} : TA \otimes TB \xrightarrow{\cong} TB \otimes TA$$

such that

$$\begin{array}{ccccc} TA \otimes TB \otimes TC & \xrightarrow{1 \otimes y} & TA \otimes TC \otimes TB & \xrightarrow{y \otimes 1} & TC \otimes TA \otimes TB \\ y \otimes 1 \downarrow & & & & \downarrow 1 \otimes y \\ TB \otimes TA \otimes TC & \xrightarrow{1 \otimes y} & TB \otimes TC \otimes TA & \xrightarrow{y \otimes 1} & TC \otimes TB \otimes TA \end{array}$$

In the case of non-strict monoidal $\mathcal{A}$, the diagram needs punctuating with six associativity isomorphisms.

b-Categories [Davydov-Runkel 2013]

A *b-category* is a category $\mathcal{A}$ equipped with a functor $L : \mathcal{A} \rightarrow [\mathcal{A}, \mathcal{A}]$ bearing a Yang-Baxter operator $\beta_{A,B,-} : LA \circ LB \xrightarrow{\cong} LB \circ LA$.

Putting $L(A)B = A \otimes B$, we obtain a magmale structure on $\mathcal{A}$ with a natural family $\beta_{A,B,C} : A \otimes (B \otimes C) \xrightarrow{\cong} B \otimes (A \otimes C)$ satisfying a hexagonal diagram whose vertices are fourfold tensors all bracketed from the right.

Each braided monoidal category $\mathcal{A}$ becomes a b-category with $\beta_{A,B,C}$:

$$A \otimes (B \otimes C) \xrightarrow{\alpha} (A \otimes B) \otimes C \xrightarrow{\gamma \otimes 1} (B \otimes A) \otimes C \xrightarrow{\alpha^{-1}} B \otimes (A \otimes C) .$$

Unital b-Categories

A b-category $\mathcal{A}$ is called *unital* when the magmale is a rumale subject to two axioms which I shall not reproduce here.

However, we see then that we have a braiding candidate $\gamma_{A,B}$:

$$A \otimes B \xrightarrow{1 \otimes \rho_B} A \otimes (B \otimes I) \xrightarrow{\beta_{A,B,I}} B \otimes (A \otimes I) \xrightarrow{1 \otimes \rho_A} B \otimes A$$

and hence an associativity candidate $\alpha_{A,B,C}$:

$$A \otimes (B \otimes C) \xrightarrow{1 \otimes \gamma^{-1}} A \otimes (C \otimes B) \xrightarrow{\beta^{-1}} C \otimes (A \otimes B) \xrightarrow{\gamma} (A \otimes B) \otimes C .$$

Proposition (Davydov-Runkel 2013)

This defines a bijection between braided monoidal structures on a rumale and unital b-category structures.

The centre of a category

- ▶ For any monoidal category $\mathcal{V}$ with tensor unit I , the endomorphism monoid $\mathcal{V}(I, I)$ is commutative. From our considerations above, we expect this since, as well as the composition binary operation, we have the convolution operation

$$(f, g) \mapsto (I \cong I \otimes I \xrightarrow{f \otimes g} I \otimes I \cong I)$$

which preserves composition.

- ▶ In particular, for any category $\mathcal{A}$, we have the strict monoidal category $[\mathcal{A}, \mathcal{A}]$ with unit $1_{\mathcal{A}}$. The commutative monoid

$$\begin{aligned} [\mathcal{A}, \mathcal{A}](1_{\mathcal{A}}, 1_{\mathcal{A}}) &= \int_A \mathcal{A}(A, A) \\ &= \{ \theta = (A \xrightarrow{\theta_A} A)_{A \in \mathcal{A}} : f \circ \theta_A = \theta_B \circ f \\ &\quad \text{for all } f \in \mathcal{A}(A, B) \} \end{aligned}$$

is called the *centre* of the category $\mathcal{A}$.

The centre of a monoid

- ▶ The centre $\mathcal{Z}M$ of a monoid M is the centre of the category ΣM with only object 0 whose endohom is $\Sigma M(0,0) = M$.
- ▶ It is the equalizer

$$\mathcal{Z}M \longrightarrow M \begin{array}{c} \xrightarrow{\partial_0} \\ \xleftarrow{\quad} \\ \xrightarrow{\partial_1} \end{array} [M, M]$$

of a reflexive pair, where $\partial_0(x) = x \cdot -$ and $\partial_1(x) = - \cdot x$.

- ▶ This allows us to define the centre of a monoid in a braided closed monoidal category as an equalizer of a reflexive pair.

The centre of a bicategory

- ▶ For any monoidal bicategory $\mathcal{M}$ with tensor unit I , the endomorphism monoidal bicategory $\mathcal{M}(I, I)$ is braided.
- ▶ In particular, for any bicategory $\mathcal{K}$, we have the strict monoidal bicategory $\text{Ps}(\mathcal{K}, \mathcal{K})$ of pseudofunctors $\mathcal{K} \rightarrow \mathcal{K}$, pseudonatural transformations, and modifications. The braided monoidal category $\text{Ps}(\mathcal{K}, \mathcal{K})(1_{\mathcal{K}}, 1_{\mathcal{K}})$ is called the *centre* of the bicategory $\mathcal{K}$.

The centre of a monoidal category

- ▶ The *centre* $\mathcal{Z}\mathcal{V}$ of a monoidal category $\mathcal{V}$ is the centre of the bicategory $\Sigma\mathcal{V}$ with only object 0 and $\Sigma\mathcal{V}(0,0) = \mathcal{V}$.
- ▶ It is the descent category of a truncated pseudo-cosimplicial category.

$$\mathcal{Z}\mathcal{V} \longrightarrow \mathcal{V} \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} [\mathcal{V}, \mathcal{V}] \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} [\mathcal{V} \otimes \mathcal{V}, \mathcal{V}]$$

- ▶ This allows us to define the centre $\mathcal{Z}V$ of any pseudomonoid (monoidale) V in a braided closed monoidal bicategory $\mathcal{M}$ as a descent construction (it is a limit of the kind appropriate to bicategories).
- ▶ When it exists, $\mathcal{Z}V$ is a braided monoidale in $\mathcal{M}$.

The use of the monoidal centre in [38]

- ▶ The goal of this paper with André Joyal was to provide a purely algebraic proof that the free tortile tensor category Tang on one generating object was also the free tensor category $\mathcal{F}$ containing an object X equipped with a tortile Yang-Baxter operator $y : X \otimes X \cong X \otimes X$.
- ▶ In her Macquarie University PhD (1989), Mei Chee Shum showed that Tang was the monoidal category of tangles on ribbons.
- ▶ We wanted to show that Tang was also monoidally equivalent to $\mathcal{F}$ purely by using the universal property of $\mathcal{F}$: strong monoidal functors $\mathcal{F} \rightarrow \mathcal{C}$ into any monoidal category $\mathcal{C}$ are essentially the same as objects of $\mathcal{C}$ equipped with a tortile Yang-Baxter operator.
- ▶ I shall outline the proof that $\mathcal{F}$ is braided.

The use (continued)

- ▶ We need to express structure in terms of strong monoidal functors to use the universal property.
- ▶ A braiding on a monoidal category $\mathcal{V}$ amounts to a strong monoidal functor $G : \mathcal{V} \rightarrow \mathcal{Z}\mathcal{V}$ whose composite with $\mathcal{Z}\mathcal{V} \rightarrow \mathcal{V}$ is $1_{\mathcal{V}}$.
- ▶ The centre of a monoidal category $\mathcal{V}$ can be described using strong monoidal functors from $\mathcal{V}$ to another construction on $\mathcal{V}$. The *centralizer* $\mathcal{C}_{\mathcal{V}}(A)$ of $A \in \mathcal{V}$ is the category whose objects are pairs $(C \in \mathcal{V}, C \otimes A \xrightarrow{u} A \otimes C)$ with the obvious morphisms and tensor product.
- ▶ Then $\mathcal{Z}\mathcal{V}$ is equivalent to the category of pairs

$$(A \in \mathcal{V}, \mathcal{V} \xrightarrow{F} \mathcal{C}_{\mathcal{V}}(A))$$

where F is a strong monoidal functor whose composite with $\mathcal{C}_{\mathcal{V}}(A) \rightarrow \mathcal{V}$ is $1_{\mathcal{V}}$; it amounts to a “half braiding” u on A .

The use (further continued)

- ▶ The generating object $X \in \mathcal{F}$ bears a tortile YB-operator $X \otimes X \xrightarrow{y \cong} X \otimes X$ which is also a tortile YB-operator on the object (X, y) of $\mathcal{C}_{\mathcal{F}}(X)$.
- ▶ By the universal property of $\mathcal{F}$, we obtain a strong monoidal functor $\mathcal{F} \xrightarrow{F} \mathcal{C}_{\mathcal{F}}(X)$ splitting $\mathcal{C}_{\mathcal{F}}(X) \rightarrow \mathcal{F}$.
- ▶ This gives an object $(X, F) \in \mathcal{Z}\mathcal{F}$ bearing the tortile YB-operator y .
- ▶ By the universal property of $\mathcal{F}$, we obtain a strong monoidal functor $\mathcal{F} \xrightarrow{F} \mathcal{Z}\mathcal{F}$ splitting $\mathcal{Z}\mathcal{F} \rightarrow \mathcal{F}$, and so the required braiding on $\mathcal{F}$.

Davydov's full centre

For a monoidal category $\mathcal{C}$ and a monoid A in $\mathcal{C}$, Davydov considers the category whose objects are maps $\zeta : Z \rightarrow A$ in $\mathcal{C}$ with (Z, u) in $\mathcal{Z}(\mathcal{C})$ and such that the following diagram commutes.

$$\begin{array}{ccc}
 Z \otimes A & \xrightarrow{\zeta \otimes 1} & A \otimes A \\
 \downarrow u_A & & \searrow \mu \\
 A \otimes Z & \xrightarrow{1 \otimes \zeta} & A \otimes A \\
 & & \nearrow \mu
 \end{array}$$

A terminal object $i : z(A) \rightarrow A$ in this category is called the *full centre* of A . Davydov proves $z(A)$ is a commutative monoid in $\mathcal{Z}(\mathcal{C})$, i is a monoid morphism in $\mathcal{C}$, and $z(A) \cong z(B)$ for Morita equivalent A and B .

Full centre as a centre

Consider the 2-category Cat_* of pairs $(\mathcal{X}, X)$ where $\mathcal{X}$ is a category and X is an object of $\mathcal{X}$, and where the morphisms $(F, \varphi) : (\mathcal{X}, X) \rightarrow (\mathcal{Y}, Y)$ consist of a functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ and a morphism $\varphi : FX \rightarrow Y$ in $\mathcal{Y}$. This is a symmetric monoidal 2-category under product.

A pseudomonoid in Cat_* is a pair $(\mathcal{C}, A)$ where $\mathcal{C}$ is a monoidal category and A is a monoid in $\mathcal{C}$. A braided pseudomonoid is a braided monoidal category $\mathcal{Z}$ with a commutative monoid Z selected.

Proposition

If the full centre $z(A)$ of a monoid A in a monoidal category $\mathcal{C}$ exists (in the sense of Davydov) then the centre of $(\mathcal{C}, A)$ in Cat_ exists and is given by*

$$\mathcal{Z}(\mathcal{C}, A) \simeq (\mathcal{Z}(\mathcal{C}), z(A)).$$

The autonomous symmetric monoidal bicategory $\mathbf{Mod}$

The objects are categories. The homcategories are defined by

$$\mathbf{Mod}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\mathrm{op}} \times \mathcal{A}, \mathbf{Set}] ;$$

objects of these homs are called *modules*. Composition

$$\mathbf{Mod}(\mathcal{B}, \mathcal{C}) \times \mathbf{Mod}(\mathcal{A}, \mathcal{B}) \xrightarrow{\circ} \mathbf{Mod}(\mathcal{A}, \mathcal{C})$$

is defined by $(N \circ M)(C, A) = \int^B M(B, A) \times N(C, B)$. The identity module $\mathcal{A} \rightarrow \mathcal{A}$ is the hom functor of $\mathcal{A}$. Tensor product is finite cartesian product of categories; it is not the product in $\mathbf{Mod}$. Coproduct in $\mathbf{Mod}$ is coproduct of categories; it is also product in $\mathbf{Mod}$. We identify each functor $F : \mathcal{A} \rightarrow \mathcal{B}$ with the module $F_* : \mathcal{A} \rightarrow \mathcal{B}$ defined by $F_*(B, A) = \mathcal{B}(B, FA)$; this gives a faithful locally-full pseudofunctor $(-)_* : \mathbf{Cat} \rightarrow \mathbf{Mod}$ which is the identity on objects, strong monoidal, and coproduct preserving. The dual of $\mathcal{A}$ in $\mathbf{Mod}$ is $\mathcal{A}^{\mathrm{op}}$.

Promonoidal categories

Monoidales in Mod^{op} are promonoidal categories $\mathcal{A}$ in the sense of Day (1970). The tensor product is a module $P : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$ and the unit is a module $J : \mathcal{A} \rightarrow \mathbf{1}$. The Day convolution monoidal structure on the functor category $[\mathcal{A}, \text{Set}]$ has unit $J : \mathcal{A} \rightarrow \text{Set}$ and tensor product

$$(F \star G)C = \int^{A,B} P(A, B; C) \times FA \times GB .$$

Suppose $\mathcal{V}$ is a complete cocomplete closed symmetric monoidal category. We have a symmetric (weak) monoidal pseudofunctor

$$[-, \mathcal{V}] : \text{Mod}^{\text{op}} \longrightarrow \mathcal{V}\text{-CAT} \quad (1)$$

taking each category $\mathcal{A}$ to the $\mathcal{V}$ -enriched functor category $[\mathcal{A}, \mathcal{V}]$. Therefore, each promonoidal category is taken to a convolution monoidal $\mathcal{V}$ -category. In the case $\mathcal{V} = \text{Set}$, (1) is the contravariant representable pseudofunctor $\text{Mod}(-, \mathbf{1})$.

Groups as one-object categories

- ▶ Let G be a group regarded as a one-object category. The object will be denoted by o_G . The morphisms of this category will also be called *elements* of G ; so $a \in G$ will mean, as usual, that a is an element.
- ▶ We use the conjugation notation ${}^c a = cac^{-1}$ and $a^c = c^{-1}ac$.
- ▶ Like every category, G has a promonoidal structure for which the convolution structure on $[G, \text{Set}]$ is pointwise cartesian product.
- ▶ In this case, the unit J is constant at the one-point set and the module $P : G \rightarrow G \times G$ is defined by

$$P(o_G, o_G; o_G) = \{(x, y) : x, y \in G\} , \quad P(a, b; c)(x, y) = (cxa, cyb) .$$

Pseudofunctors from group morphisms

- ▶ A group morphism $\pi : H \rightarrow G$ is a functor.
- ▶ Any functor into a category G corresponds to a normal lax functor $\mathbf{H} : G \rightarrow \text{Mod}^{\text{op}}$ [Bénabou].
- ▶ The functor π is a fibration if and only if it is an opfibration if and only if it is Giraud-Conduché if and only if it is a surjective group morphism. In this case, the normal lax functor is actually a pseudofunctor which we now describe.
- ▶ The fibre of the functor π over o_G is the kernel K of the group morphism π ; so $\mathbf{H}o_G = K$ as a one-object category. For $a : o_G \rightarrow o_G$ in G , the module $\mathbf{H}a : K \rightarrow K$ is defined by

$$(\mathbf{H}a)(o_K, o_K) = \{x \in H : \pi(x) = a\} .$$

In particular, $\mathbf{H}1 = K$ is the identity module.

Pseudofunctors from group morphisms, continued

- ▶ For a composable pair $o_G \xrightarrow{b} o_G \xrightarrow{a} o_G$ in G , the composite module $\mathbf{H}b \circ \mathbf{H}a$ is seen to be the set

$$(\mathbf{H}b \circ \mathbf{H}a)(o_K, o_K) = (\{x \in H : \pi(x) = a\} \times \{y \in H : \pi(y) = b\}) / K$$

of orbits under the action of K on the right in the first factor of the product and on the left in the second.

- ▶ The composition constraint is the bijection

$$(\mathbf{H}b \circ \mathbf{H}a)(o_K, o_K) \longrightarrow \mathbf{H}(ab)(o_K, o_K) , \quad [x, y] \mapsto xy .$$

- ▶ $\mathbf{H}$ is a monoidale in $\mathbf{Ps}(G, \mathbf{Mod}^{\text{op}})$. We have $I = !_* : \mathbf{H}o_G \rightarrow \mathbf{1}$ and $\square : \mathbf{H}o_G \rightarrow \mathbf{H}o_G \times \mathbf{H}o_G$ defined by

$$\square(o_K, o_K; o_K) = \{(u, v) : u, v \in K\} , \quad \square(k, \ell; m)(u, v) = (muk, mv\ell) .$$

The example of G -sets

The groupoid of automorphisms in G will be denoted by G^{aut} . The objects are elements $a \in G$. A morphism $f : a \rightarrow b$ is an element $f \in G$ such that $fa = bf$. If G_{conj} denotes the G -set whose elements are those of G and whose G -action is by conjugation then there is an equivalence of categories

$$[G, \text{Set}] / G_{\text{conj}} \simeq [G^{\text{aut}}, \text{Set}] . \quad (2)$$

Since G_{conj} is a monoid in $[G, \text{Set}]$, there is a monoidal structure on the left hand side of (2) whose tensor product takes cartesian product of the morphisms over G_{conj} followed by the monoid multiplication. This monoidal structure is closed and so transports to a promonoidal structure on G^{aut} . This promonoidal structure is defined by

$$P(a, b; c) = \{(u, v) \in G \times G : {}^u a {}^v b = c\} \quad \text{and} \quad Jc = \begin{cases} 1 & \text{if } c = 1 \\ \emptyset & \text{if } c \neq 1 \end{cases}$$

and there is a braiding γ and twist τ :

$$\gamma_{a,b;c} : P(a, b; c) \xrightarrow{\cong} P(b, a; c) , \quad (u, v) \mapsto ({}^u a v, u) , \quad \tau_a = a : a \rightarrow a .$$

The example of G -sets, continued

Furthermore, G^{aut} is a $*$ -autonomous promonoidal category: we have the natural isomorphisms

$$P(a, b; c^{-1}) \xrightarrow{\cong} P(b, c; a^{-1}) , \quad (u, v) \leftrightsquigarrow (u^{-1}v, u^{-1}) ;$$

$$P(a, b; c^{-1}) \xrightarrow{\cong} P(b^{-1}, a^{-1}; c)^{\text{op}} , \quad (u, v) \leftrightsquigarrow (v, u) .$$

Proposition

The braided monoidal G^{aut} is the monoidal centre of the monoidal G in Mod^{op} . The convolution braided monoidal category $[G^{\text{aut}}, \text{Set}]$ is braided monoidal equivalent to the monoidal centre $\mathcal{Z}[G, \text{Set}]$ of the cartesian monoidal category $[G, \text{Set}]$ of G -sets.

We note that the equivalence becomes balanced on transport of the convolution twist to the monoidal centre.

The lifting $\mathbf{H}^{\text{aut}}$ of $\mathbf{H}$ to G^{aut}

- ▶ We return to our surjective group morphism $\pi : H \rightarrow G$. Since fibrations in $\mathbf{Cat}$ are preserved by 2-functors of the form $[\mathcal{D}, -]$, we have the fibration $\pi^{\text{aut}} : H^{\text{aut}} \rightarrow G^{\text{aut}}$.
- ▶ The corresponding pseudofunctor $\mathbf{H}^{\text{aut}} : G^{\text{aut}} \rightarrow \mathbf{Mod}^{\text{op}}$ is defined as follows.
- ▶ For each $a \in G$, the category $\mathbf{H}^{\text{aut}} a$ has objects those $x \in H$ with $\pi(x) = a$; morphisms $k : x \rightarrow x_1$ are those $k \in K$ with $kx = x_1$; composition is that of K .
- ▶ For $f : b \rightarrow a$ in G^{aut} , the module $\mathbf{H}^{\text{aut}} f : \mathbf{H}^{\text{aut}} a \rightarrow \mathbf{H}^{\text{aut}} b$ is defined by $(\mathbf{H}^{\text{aut}} f)(y, x) = \{h \in H^{\text{aut}}(y, x) : \pi(h) = f\}$.

The lifting $\mathbf{H}^{\text{aut}}$ of $\mathbf{H}$ to G^{aut} , continued

- $\mathbf{H}^{\text{aut}}$ is a balanced monoidale in $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$. The monoidal structure is provided by the modules

$$\square_{a,b} : \mathbf{H}^{\text{aut}}(ab) \rightarrow \mathbf{H}^{\text{aut}} a \times \mathbf{H}^{\text{aut}} b \quad \text{and} \quad l : \mathbf{H}^{\text{aut}} 1 \rightarrow 1$$

defined by

$$\square_{a,b}(x, y; z) = \{(u, v) : u, v \in K, {}^u x {}^v y = z\} \quad \text{and} \quad lz = \begin{cases} 1 & \text{if } z = 1 \\ \emptyset & \text{if } z \neq 1 \end{cases}$$

The braiding is

$$\gamma_{x,y;z} : \square_{a,b}(x, y; z) \xrightarrow{\cong} \square_{b,a}(y, x; z), \quad (u, v) \mapsto ({}^u x {}^v y, u).$$

The twist $\tau : \mathbf{H}^{\text{aut}} \rightarrow \mathbf{H}^{\text{aut}}$ is given by $\tau_a = \mathbf{H}^{\text{aut}} a : \mathbf{H}^{\text{aut}} a \rightarrow \mathbf{H}^{\text{aut}} a$.

A higher full centre

- ▶ We can lift the concept of full centre from the monoidal category level to the monoidal bicategory level.
- ▶ The full centre of a monoidale $\mathbb{M}$ in a monoidal bicategory $\mathcal{K}$ is a braided monoidale $z\mathbb{M}$ in the monoidal centre $\mathcal{Z}\mathcal{K}$ of $\mathcal{K}$ satisfying an appropriate universal property.

Proposition

- (i) *The braided monoidal bicategory $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$ is the monoidal centre of the monoidal bicategory $\text{Ps}(G, \text{Mod}^{\text{op}})$.*
- (ii) *The full centre of the monoidale $\mathbf{H}$ in $\text{Ps}(G, \text{Mod}^{\text{op}})$ is the braided monoidale $\mathbf{H}^{\text{aut}}$ in $\text{Ps}(G^{\text{aut}}, \text{Mod}^{\text{op}})$.*

Using (1) the pseudofunctor $\text{Mod}^{\text{op}} \xrightarrow{[-, \mathcal{V}]} \mathcal{V}\text{-CAT}$ with $\mathcal{V} = \text{Vect}$, we refind results about centres of the G -categories of Turaev-Virelizier [JPAA, 2013].

Thank you for inviting me and for your attention!

